



StreamIQ

SQL PROJECT

StreamIQ

SQL Portfolio Project

From relational data to business insights across acquisition, engagement, content, revenue, retention and churn — built on MySQL.

10,000

Users

2,000

Content Titles

200,000

Viewing Sessions

60,000

Payments

Dataset: synthetic / fictional StreamIQ dataset created for portfolio analysis — not real company data.

Use SQL to answer realistic product and business questions across the customer lifecycle — from acquisition and viewing behavior to subscriptions, payments and cancellations.

Analysis areas

01

User & Acquisition

Geography, signup growth, account status, age groups and channel mix

02

Engagement

Viewing volume, active viewers, discovery sources, devices, completion

03

Content

Top titles, watch time, genres, languages, unwatched catalog

04

Monetization

Active-subscription MRR, successful revenue, payment-method risk

05

Retention

30/60/90-day viewing retention by cohort and user segment

06

Churn

Cancellations by plan, auto-renewal and acquisition channel

07

Advanced Analytics

Running totals, NTILE, RANK, CTEs, subscriber-value analysis

The Dataset



users

Customer profile, geography, signup & acquisition



content

Streaming catalog — type, language, rating, runtime



genres

Genre lookup table



content_genres

Many-to-many bridge: content ↔ genres



devices

Registered devices and operating systems



subscriptions

Plans, prices, lifecycle dates, status, auto-renew



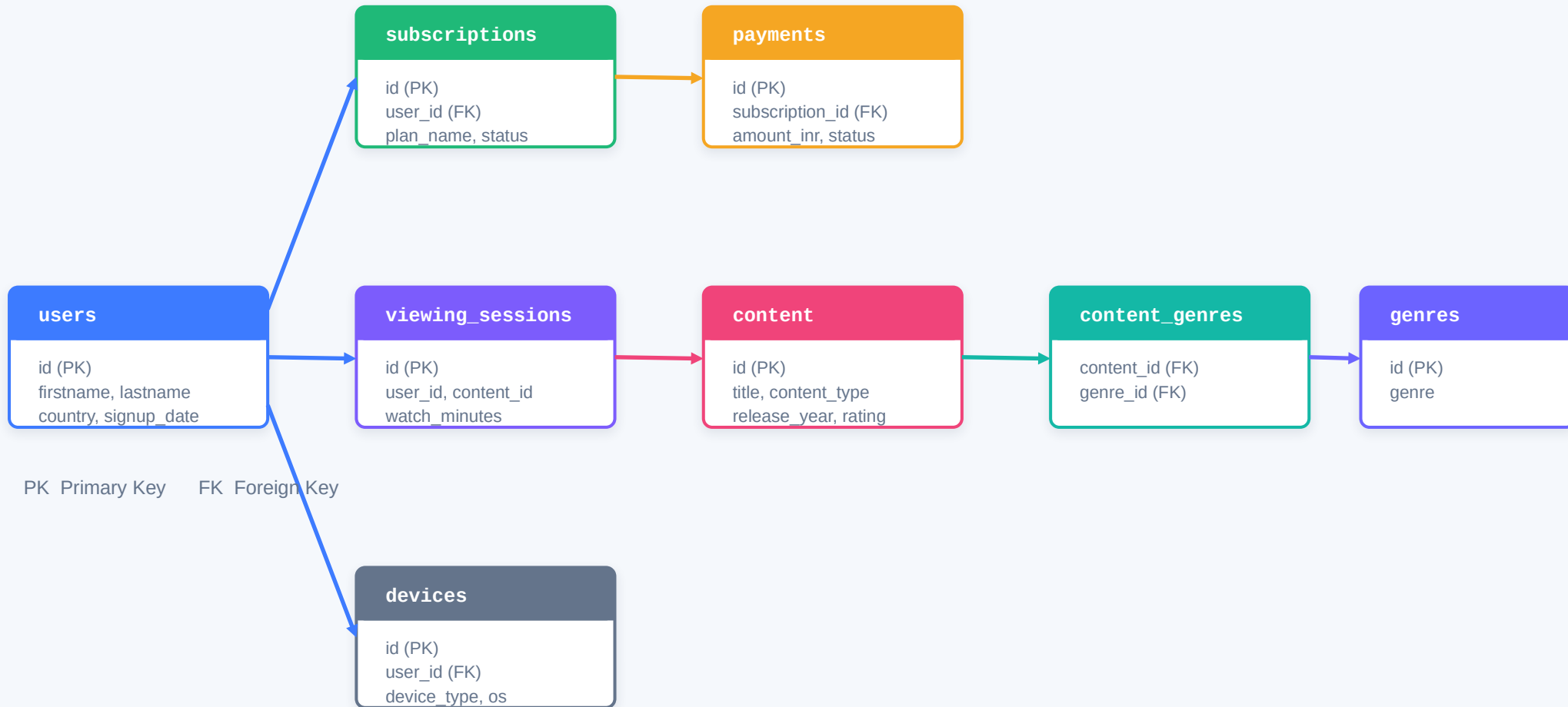
payments

Payment history — amounts, methods, statuses



viewing_sessions

Watch behavior, source, quality, completion



10,000

Users

10

Countries

2,000

Content Titles

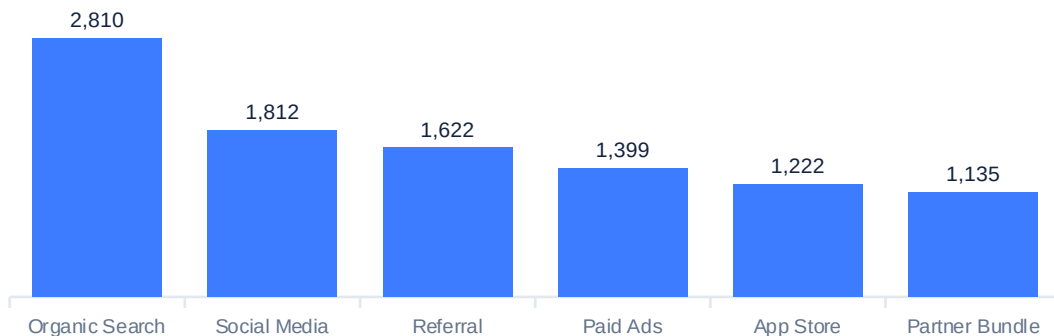
200,000

Viewing Sessions

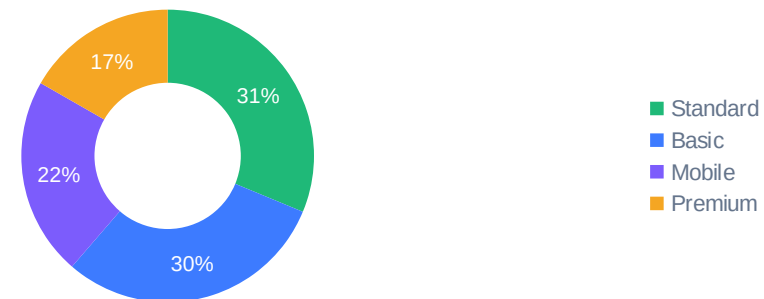
60,000

Payments

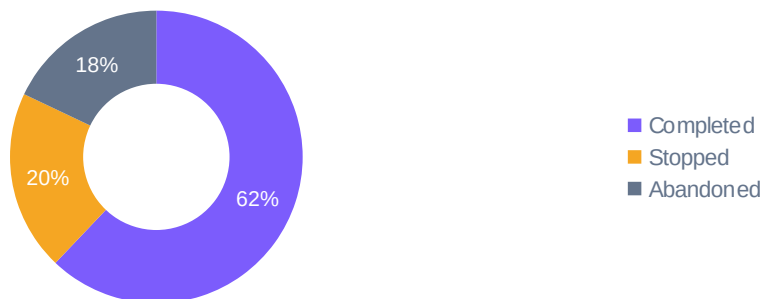
User Acquisition Channels



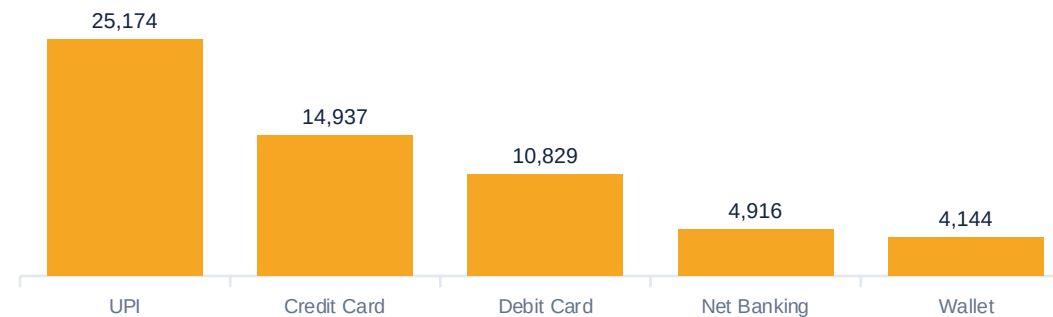
Subscription Plan Distribution



Viewing Session Status



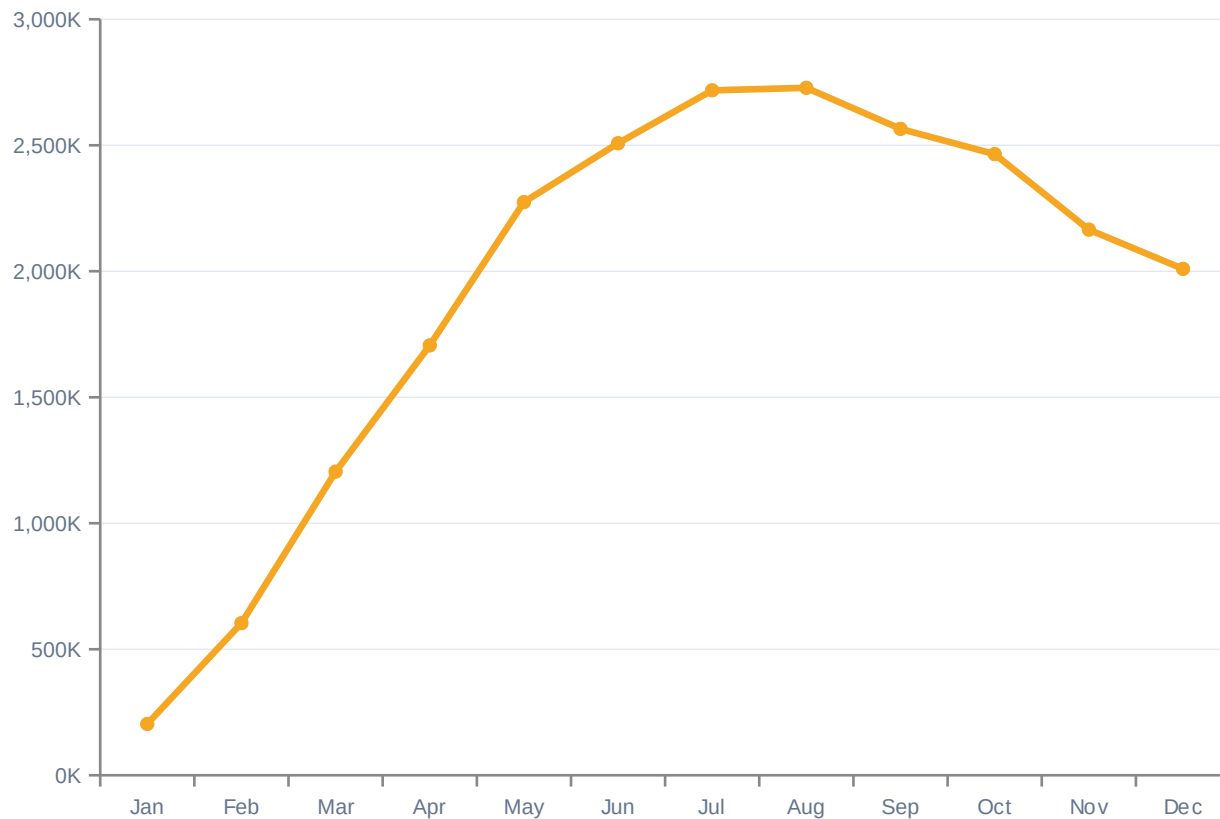
Payment Method Usage



Key takeaway: Organic Search brings in the most users (2,810); UPI is the most-used payment method (25,174 attempts); and 62.1% of sessions run to completion.

Monthly Revenue Trend (INR, Successful Payments)

Peak: Aug 2025 · ₹27.3L



Revenue climbs steadily from Jan through Aug 2025 as active subscriptions build up, then eases through Q4 as cancellations outpace new signups late in the year.

SQL Analysis Roadmap

1

Platform & User Overview

Aggregation, percentages, date grouping, segmentation

2

User Engagement

Joins, monthly activity, device & source breakdowns

3

Content Performance

Multi-table joins, genre bridge, ranking, LEFT JOIN

4

Subscription & Revenue

MRR proxy, successful revenue, payment-method risk

5

Retention & Lifecycle

Conditional logic, cohort-style 30/60/90-day retention

6

Churn & Subscription

Cancellation rates, duration analysis, partitioned RANK

7

Advanced Product Analytics

Running totals, NTILE, RANK, CTES, subscriber value

1

CTE — Monthly Cohort Retention

Groups users into signup-month cohorts, then measures what share viewed content within 30 days.

```
WITH user_retention AS (  
  SELECT u.user_id,  
    DATE_FORMAT(u.signup_date, '%Y-%m') cohort,  
    MAX(CASE WHEN v.started_at <  
      DATE_ADD(u.signup_date, INTERVAL 30 DAY)  
    THEN 1 ELSE 0 END) retained  
  FROM users u LEFT JOIN viewing_sessions v  
    ON u.user_id = v.user_id  
  GROUP BY u.user_id, cohort)  
SELECT cohort, AVG(retained) FROM user_retention  
GROUP BY cohort;
```

2

Window Function — LAG()

Compares each month's new-signup count to the prior month to surface acceleration or slowdown.

```
SELECT signup_month, new_users,  
  LAG(new_users) OVER (  
    ORDER BY signup_month  
  ) AS previous_month,  
  new_users - LAG(new_users) OVER (  
    ORDER BY signup_month  
  ) AS user_change  
FROM monthly_users;
```

3

RANK() — Subscriber Engagement

Within each subscription plan, ranks active subscribers by total watch time to find top users.

```
WITH ranked AS (  
  SELECT user_id, plan_name,  
    total_watch_minutes,  
    RANK() OVER (  
      PARTITION BY plan_name  
      ORDER BY total_watch_minutes DESC  
    ) AS plan_rank  
  FROM subscriber_engagement)  
SELECT * FROM ranked  
WHERE plan_rank <= 10;
```

4

Complex Join — Genre Popularity

Three-table join through the content_genres bridge table to roll viewing sessions up to genre level.

```
SELECT g.genre,  
  COUNT(v.session_id) AS sessions,  
  SUM(v.watch_minutes) AS watch_minutes  
FROM genres g  
JOIN content_genres cg ON g.genre_id = cg.genre_id  
JOIN viewing_sessions v ON cg.content_id = v.content_id  
GROUP BY g.genre  
ORDER BY watch_minutes DESC;
```

Key Business Insights

- Organic Search is the top acquisition channel (2,810 users, 28% of the base).
- India and the US lead the user base by country — 34% and 18% respectively.
- UPI is the most-used payment method (25,174 of 60,000 transactions).
- Payment success rate is 92.98% — the remaining ~7% represents lost revenue.
- 30-day retention sits at 80.3%; most users return within their first month.
- Churn is 23.3% and nearly flat across every plan (22.9%–23.7%).
- 252 of 2,000 catalog titles (12.6%) have never been watched.

Data Caveats & Honest Interpretation

Synthetic data: StreamIQ is fictional, built for portfolio practice — not real company data.

Payments: 60,000 payment records are a capped history, not an exhaustive ledger.

Retention: 30/60/90-day metrics track viewing activity after signup, not formal subscription retention.

Churn: Cancellation analysis is subscription-record based, not unique-user churn.

MRR & ARPU: MRR is an active-subscription proxy; ARPU is computed across available payment history.

SQL Techniques Used

JOINS

GROUP BY

CASE WHEN

CTEs

Window Functions

RANK()

NTILE()

LAG()

HAVING

Subqueries



StreamIQ

Thank You!

MySQL · MySQL Workbench · SQL Analytics



Tools & Technologies

What was used to build and analyze StreamIQ

8/9



MySQL

Core database engine for all CREATE, INSERT and SELECT operations



MySQL Workbench

GUI tool used for schema design, ER diagram and query execution



GitHub

Version control and hosting for all SQL files and documentation



PowerPoint

Presentation of findings, schema diagrams and query explanations



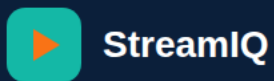
SQL Concepts

SELECT, WHERE, JOIN, GROUP BY, HAVING, Window Functions, CTEs



Database Design

Normalization, primary keys, foreign keys and relational schema



Thank You!

Questions & Discussion

📁 **GitHub Repo:** <https://github.com/parihatrahul-ops/streamiq-sql-analytics>

🔧 **Tech Stack:** MySQL · MySQL Workbench · GitHub

User & Acquisition

Engagement

Content

Monetization

Retention

Churn

Advanced Analytics